Aaron Isac Esquivel San Juan

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EDUCATION

Master of Science in Electrical Engineering Jan 2026 – Present
California State University, Los Angeles | GPA: 3.57/4.0 | Expected graduation: Fall 2027

Bachelor of Science in Mechatronics Engineering May 2025
Centro de Enseñanza Técnica y Superior (CETYS) | GPA: 3.9/4.0 (WES evaluated)

Bachelor of Arts in Management Jun 2025
City University of Seattle | GPA: 3.6

ENGINEERING EXPERIENCE

Technical Inspector | Diesel Power Systems | Jul 2025 - Dec 2025

- Reviewed drawings and inspected electrical and gas installations in homes, businesses, and industrial facilities, including SILZA plants in northwestern Mexico, for safety and compliance.
- Prepared technical drawings and inspection documentation, used ultrasonic equipment to detect leaks, and reported findings for technical review.

R&D Intern - Circuits and Components | Navico Group | Oct 2024 - Feb 2025

- Tested marine LCD components with optical instrumentation and supported hardware performance and glass-coating evaluations.
- Developed a Python interface for an LCD optical measurement instrument, automating tests and data collection.
- Designed and 3D-printed test and prototype parts; documented test procedures and results for the R&D team.

Research Intern - Machine Learning and Pattern Recognition | CICESE | Mar 2024 - Jul 2024

- Collaborated with a doctoral researcher to collect and label gyroscope data and develop Python machine learning algorithms that identify human activities from movement.

ENGINEERING PROJECTS

PillDrop Automated Medication and Water Dispenser

- Built a medication and water dispenser using Arduino and ESP32 for people with reduced mobility. Integrated Alexa voice control, designed the PCB in Proteus 8, and 3D-printed parts. Team won 1st place at the CETYS 2023 Engineering Expo.

Autonomous Drone

- Implemented and tested trajectory-tracking algorithms in MATLAB and Simulink, with Bluetooth for real-time control. Team placed 2nd in the MathWorks National Drone Competition finals in Mexico City.

E-Serve Autonomous Food and Water Delivery Robot

- Built an autonomous delivery robot using Arduino, Raspberry Pi, and ROS, with LiDAR obstacle detection and 3D-printed and CNC-machined parts. Won 2nd place at the CETYS 2022 Engineering Expo.

LEADERSHIP

Founding Member and President | CETYS Ensenada Robotics Club | Oct 2022 - Jun 2025

- Organized the team, coordinated robotics projects and activities, and identified competition opportunities for the club.
- Built autonomous competition robots with sensor and actuator integration, PCB design in Proteus 8, and PID control.

Vice President | CETYS Engineering Student Council | Aug 2024 - May 2025

- Worked with council members to organize student activities and coordinate guest speakers for engineering talks.

TECHNICAL SKILLS

Programming: C/C++, Python; Linux, Git, GitHub.
Modeling and simulation: MATLAB, Simulink, FluidSIM.
Embedded systems and control: Arduino, ESP32, Raspberry Pi, Zephyr RTOS, ROS, PID control, sensor integration.
Testing and instrumentation: Hardware and circuit testing, component validation, test automation, data collection and analysis; oscilloscopes and logic analyzers (academic labs).
Electrical and mechanical design: Proteus 8, SolidWorks, AutoCAD.
Industrial automation: Siemens PLC programming, ladder logic, TIA Portal.
Prototyping and fabrication: 3D printing, CNC machining, milling, lathe operation, laser cutting, welding, soldering.
Languages: Spanish (native), English (C1), German (A1).